

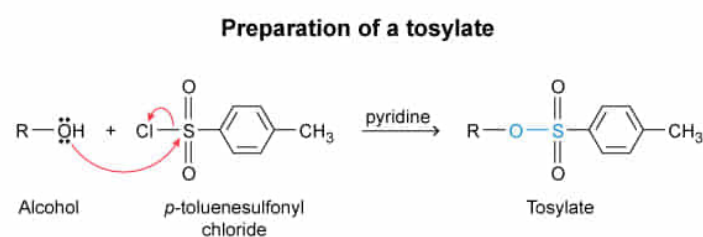
During the synthesis of a tosylate, which of the following pairs of atoms form a covalent bond?

- ☐ A. C and Cl [4%]
✓ ☒ B. S and O [67%]
☐ C. S and Cl [15%]
☐ D. C and O [12%]

Correct

66% answered correctly

Explanation:



A bond between O and S is formed during the synthesis of a tosylate.

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Substitution reactions require strong **nucleophiles and electrophiles**, as well as good **leaving groups**. The nucleophile attacks the electrophile and displaces the leaving group. A bond is formed between the nucleophile and electrophile, whereas the bond between the electrophile and the leaving group is broken.

Alcohols are weak electrophiles and **poor leaving groups**; therefore, a nucleophile is unlikely to displace the hydroxyl group ($-\text{OH}$) of an alcohol. To **improve its leaving group ability**, an alcohol can be converted into a **tosylate** ($-\text{SO}_3\text{C}_6\text{H}_4\text{CH}_3$) or a **mesylate** ($-\text{SO}_3\text{CH}_3$).

Tosylates are **produced by** the reaction of an alcohol with ***p*-toluenesulfonyl chloride (TsCl)** and a **base**, such as pyridine. The hydroxyl **O** acts as the nucleophile to **attack** the **S** of TsCl, and Cl acts as the leaving group. Then pyridine deprotonates the O atom, giving the tosylate product.

Therefore, a bond between S and O is formed during the synthesis of a tosylate.

(Choice A) No C–Cl bond is present at any point in the formation of a tosylate.

(Choice C) An S–Cl bond is present in *p*-toluenesulfonyl chloride and is *broken*, rather than formed, during the formation of a tosylate.

(Choice D) A C–O bond is present in the alcohol starting material. It is *not* formed in the reaction.

Educational objective:

Alcohols are poor leaving groups and weak electrophiles; therefore, nucleophiles are unlikely to displace the hydroxyl (–OH) groups of alcohols in substitution reactions. To improve its leaving group ability, the –OH can be converted into a tosylate (–SO₃C₆H₄CH₃) or a mesylate (–SO₃CH₃) through a reaction with a base and *p*-toluenesulfonyl chloride (TsCl) or methanesulfonyl chloride (MsCl), respectively. The hydroxyl oxygen attacks the sulfur of TsCl or MsCl, forming a bond between S and O. Cl acts as the leaving group, breaking the bond between S and Cl.

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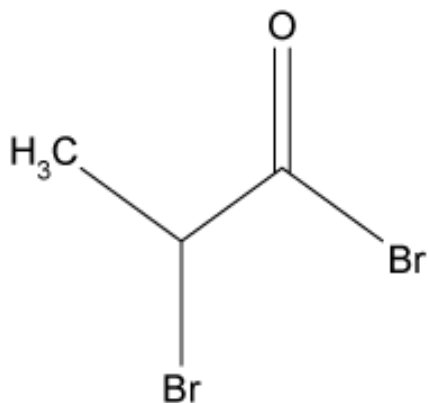
Subject:
Organic Chemistry

QID:402763

System:
Functional Groups and Their Reactions

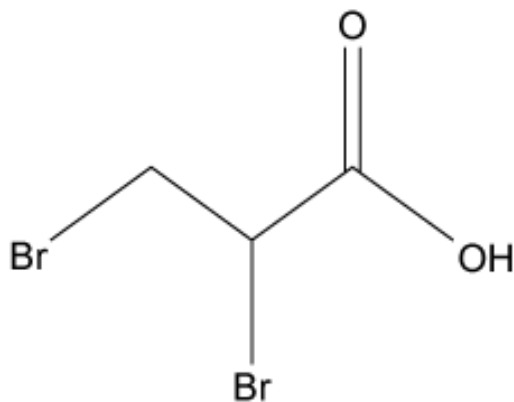
When a carboxylic acid reacts with phosphorus tribromide and bromine and then water, an α -bromoacid is formed. What is the structure of the product of this reaction if a three-carbon carboxylic acid is used as the reactant?

☐ A.



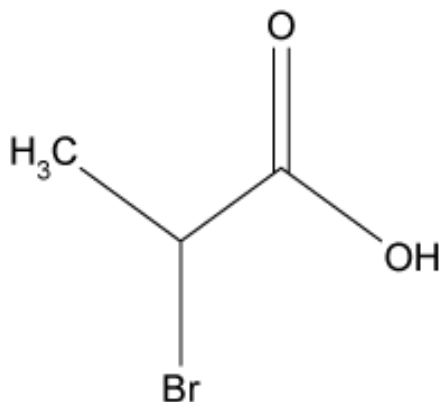
[18%]

☐ B.



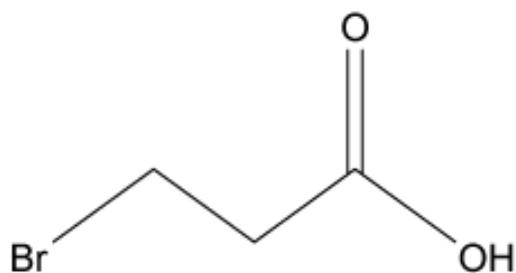
[13%]

☒ C.



[63%]

☐ D.

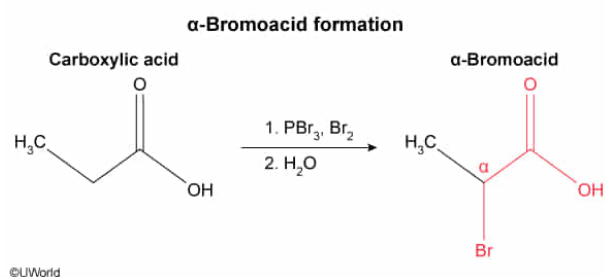


[4%]

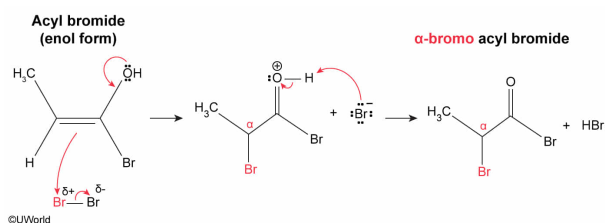
Correct

63% answered correctly

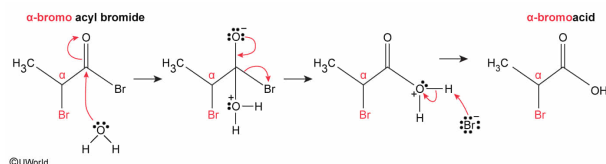
Explanation:



The question indicates that when carboxylic acids react with phosphorus tribromide (PBr_3), bromine (Br_2), and then water, bromination occurs at the **α -carbon** (carbon atom adjacent to a carbonyl carbon), forming an **α -bromoacid**. The α -bromoacid can be used to [synthesize \$\alpha\$ -amino acids](#). The bromination reaction begins with the [addition of \$\text{PBr}_3\$](#) to the carboxylic acid, forming an acyl bromide that can [tautomerize](#) from the keto to the enol form. The enol form then acts as a nucleophile and attacks one of the bromine atoms in Br_2 , adding bromine to the α -carbon and forming an α -bromo acyl bromide.



The final step is a [hydrolysis reaction](#) in which water acts as a nucleophile and attacks the carbonyl carbon, and the bromine on the carbonyl carbon acts as the leaving group. Bromide deprotonates the water substituent on the carbonyl carbon, forming the α -bromoacid.



The question states that an α -bromoacid was formed from the reaction of a three-carbon carboxylic acid with PBr_3 , Br_2 , and then water. Therefore, the product must be a three-carbon carboxylic acid with a bromine on the α -carbon only. **Choice C** is the only structure that shows such a compound.

(Choice A) This structure is the α -bromo acyl bromide intermediate formed after addition of PBr_3 and Br_2 to the carboxylic acid.

(Choice B) This carboxylic acid contains bromine atoms on both the α - and β -carbons rather than the α -carbon only.

(Choice D) This carboxylic acid is a β -bromoacid, meaning bromine is on the β -carbon (two carbon atoms away from the carbonyl carbon) rather than the α -carbon.

Educational objective:

The α -carbon of a carboxylic acid is the carbon adjacent to the carbonyl carbon and can undergo substitution reactions to form α -substituted carboxylic acids. An α -bromoacid is a carboxylic acid with bromine on the α -carbon and is formed when a carboxylic acid is reacted with PBr_3 , Br_2 , and then H_2O .

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Subject:

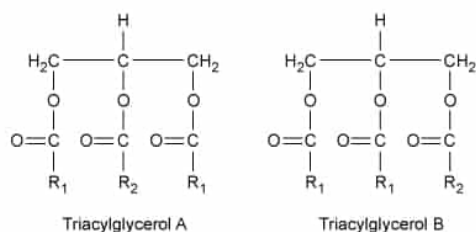
Organic Chemistry

QID: 402764

System:

Functional Groups and Their Reactions

Triacylglycerols A and B are shown below.



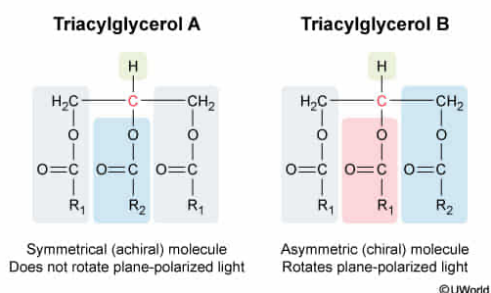
Which of the triacylglycerols would rotate plane-polarized light?

- ☐ A. Both triacylglycerols [10%]
- ☐ B. Neither triacylglycerol [26%]
- ☐ C. Triacylglycerol A only [10%]
- ✓ ☒ D. Triacylglycerol B only [53%]

Correct

53% answered correctly

Explanation:



A **chiral molecule** contains at least one atom (typically carbon) that has **four different substituents**. This atom is known as a **stereocenter**, or **chiral center**. If the central atom is bonded to two or more of the same atoms, then the subsequent atoms are compared to determine whether the substituents are different. This process is repeated until a difference is found. If no difference is found in the two substituents, the central atom is not a stereocenter.

Plane-polarized light is generated when normal light (waves oscillating in all directions) passes through a polarizer. The polarizer allows only light that oscillates in a particular direction to pass through it. Chiral molecules **interact** with plane-polarized light, causing the angle of oscillation to rotate.

Triacylglycerols are lipids made up of three fatty acids attached to a glycerol molecule through **ester** linkages. All three fatty acids may be identical, two of the three may be identical to each other, or all three may be distinct. If at least one fatty acid is different from the other two, then the middle carbon may be a stereocenter. Both Triacylglycerol A and Triacylglycerol B have two identical fatty acids and one distinct one, but Triacylglycerol A is **arranged symmetrically**, so it is **not chiral** and will not rotate plane-polarized light (**Choice C**). Triacylglycerol B is **not arranged symmetrically**, so it is **chiral** and will rotate plane-polarized light.

(Choices A and B) Only Triacylglycerol B has a stereocenter; therefore only it, and *not* Triacylglycerol A, *will* rotate plane-polarized light.

Educational objective:

Chiral molecules contain at least one atom that has four different substituents, known as a stereocenter, or chiral center. If the atom is bonded to two or more of the same atom, the subsequent atoms are

compared to determine if the substituents are different. Chiral molecules can interact with and rotate plane-polarized light.

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